ADVANCED PROBLEMS FOR LECTURE 16

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

Definition 0.1. A *partition* P of $[a, b]$ is a finite set $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$.

The *upper sum* corresponding to a function f and a partition P of $[a, b]$ is the quantity:

$$U(f, P) = \sum_{k=1}^{n} f(x_i^*)(x_i - x_{i-1})$$

where x_i^* is the element of $[x_{i-1}, x_i]$ that maximizes f over $[x_{i-1}, x_i]$

The *lower sum* corresponding to a function f and a partition P of $[a, b]$ is the quantity:

$$L(f, P) = \sum_{k=1}^{n} f(x_{i*})(x_i - x_{i-1})$$

where x_{i*} is the element of $[x_{i-1}, x_i]$ that minimizes f over $[x_{i-1}, x_i]$

We say that a function f is *integrable* over $[a, b]$ if there is a unique real number I such that

$$L(f, P) \leq I \leq U(f, P)$$

holds for all partitions P of $[a, b]$. In this case, we say that *the (definite) integral* of f over $[a, b]$ is the quantity

$$\int_a^b f(x)dx = I$$

(1) We say a partition P_1 is *finer* than P_2, and write $P_1 \leq P_2$ if $P_2 \subseteq P_1$. The idea here is that our partition points chop up our interval, so if P_1 has more chopping points then the interval is chopped "finer". Show that if $P_1 \leq P_2$ then

$$L(f, P_2) \leq L(f, P_1) \leq U(f, P_1) \leq U(f, P_2).$$

That is, upper sums get smaller under refinement, and lower sums get larger.

(2) Suppose f is continuous on $[a, b]$ (hence integrable). Find a sequence of partitions, P_n, that gets strictly increasingly finer but such that $U(f, P_n)$ does not converge to the integral of f over $[a, b]$. What condition should we place on P_n to ensure convergence?

(3) Show that the Dirichlet function is not (Riemann) integrable*. Reminder: the Dirichlet function is the function

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}.$$

(4) Is it possible for an unbounded function to be integrable?

* I have "Riemann" in parentheses here, as this integral is named after the mathematician Bernhard Riemann. However, there is a more robust (but more complicated to describe) integral named after the mathematician Henri Lebesgue. The Dirichlet function is Lebesgue-integrable, but not Riemann-integrable. These two integrals agree for every Riemann-integrable function, but there are more functions that are Lebesgue-integrable. That being said, the Riemann approach is often more practical to use in other contexts, so it tends to show up more often.